

SAYAN GOSWAMI

MSc in Data Science and AI, RKMVERI, Belur

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SUMMARY

My experience combines a strong foundation in Machine learning, Deep learning, Multimodal AI & Retrieval-Augmented Generation(RAG's) with hands-on expertise in building end-to-end intelligent systems. I have worked on projects spanning healthcare AI, predictive analytics, Generative AI, and energy optimization, and LLM evaluation, covering data preprocessing, model development, deployment, and performance assessment. My current research focuses on multimodal learning for chest X-ray analysis and medical imaging, while my broader interests include Gen AI, Retrieval-augmented systems(RAG's), and real-world AI applications.

EDUCATION

Bankura Christian College, Bankura
BSc (Honours) in Mathematics, 2022–25
CGPA –7.72

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science and Artificial Intelligence, 2025–27

EXPERIENCE

Research Intern - Indian Institute of Technology, Patna

Supervisor: Dr. Joydeep Chandra
May 2026 – Present

Working on a label-wise reliability-aware framework for robust chest X-ray multi-label classification on the NIH Chest X-rays (ChestX-ray14) dataset, using medical description prompts, GMM-based noisy-label stratification, and graph-based label refinement to improve robustness on tail classes.

SKILLS

- **Languages:** Python, C, R Programming, LaTeX
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, Langchain, PyTorch, Matplotlib
- **Operating Systems:** macOS, Ubuntu Linux
- **Tools:** Git/GitHub, Jupyter Notebook, Google Colab, MS Office

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, LLM's and Agentic AI, Blockchain
- Probability and Statistics, Linear Algebra, Time Series Analysis
- Distributed Computing, Data Structures and Algorithms

Achievements & Certificates

- NPTEL – The Joy of Computing using Python (2026) ✓
- CUET-PG 2025 (Mathematics) – Qualified

PROJECTS

NādaAI - Deep Learning Based Music Generation

RKMVERI

Jan 2026 – Apr 2026

Developed an end-to-end deep learning pipeline that converts 5 seconds of vocal humming into structured musical melodies. Integrated CREPE-based pitch detection, MIDI sequence generation, and a two-layer LSTM architecture trained on approximately 495,000 musical note sequences. Implemented real-time audio synthesis and a Streamlit interface, enabling melody generation without requiring MIDI input or formal music knowledge.

Customer Churn Prediction Using Machine Learning

RKMVERI

Oct 2025 – Dec 2025

Developed an end-to-end machine learning pipeline to predict telecom customer churn using demographic, billing, and service usage data from 7,043 customers. Implemented preprocessing, SMOTE-based class balancing, model selection, hyperparameter optimization, and performance evaluation. Deployed the final tuned Random Forest model as an interactive Streamlit dashboard providing churn risk prediction, feature importance analysis, and business decision support.

RAG Evaluation & LLMOps Pipeline

RKMVERI

Feb 2026 – May 2026

Built and evaluated a Retrieval-Augmented Generation (RAG) pipeline (Mistral + FAISS + LangChain) in two production-style configurations — as a monitored GenAI system and as an AIOps "IT Copilot" — with automated RAGAS-based quality scoring and a CI/CD quality gate via pytest and GitHub Actions.

PriceScope - Distributed Price Comparison Platform

RKMVERI

Jan 2026 – May 2026

Built a distributed price comparison platform that aggregates product information from multiple e-commerce sources and enables real-time price comparison. Implemented data collection, preprocessing, and search functionalities to support efficient cross-platform product discovery and analysis.

Smart HVAC - Deep RL Energy Optimization

RKMVERI

Aug 2025 – Dec 2025

Developed an AI-powered HVAC optimization system using Deep Reinforcement Learning (DQN), physics-based thermal simulation, and a real-time digital twin dashboard. Built a custom Gymnasium environment with thermodynamic modeling, adaptive comfort prediction, and autonomous energy-efficient control using Stable-Baselines3 and Streamlit.

ACADEMIC RESEARCH

Mathematical Modeling Using Calculus

B.Sc. Dissertation

Feb 2025 – Jun 2025

Studied mathematical modeling of dynamic systems using calculus, focusing on optimization, logistic population growth, epidemic modeling, and engineering applications.